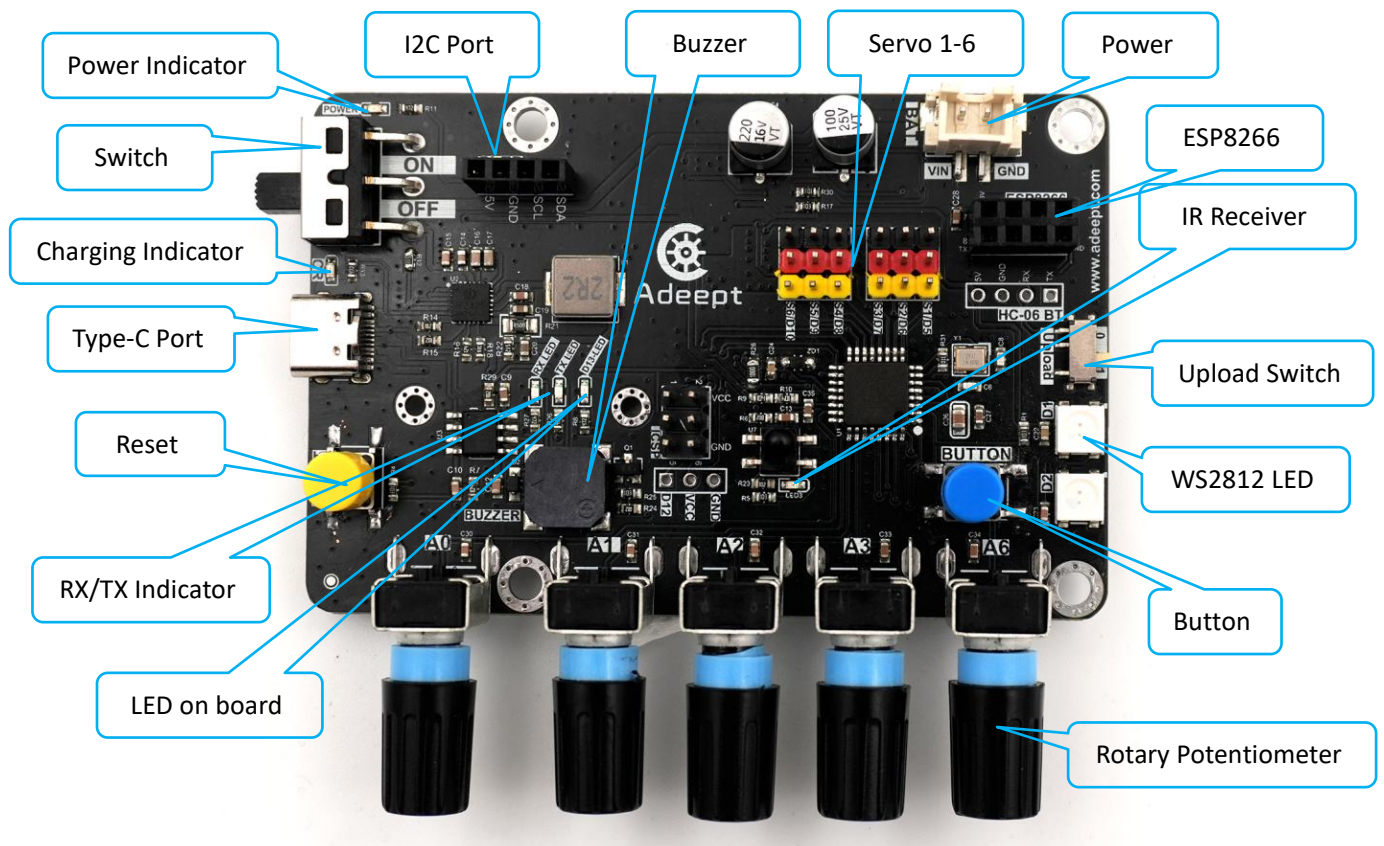
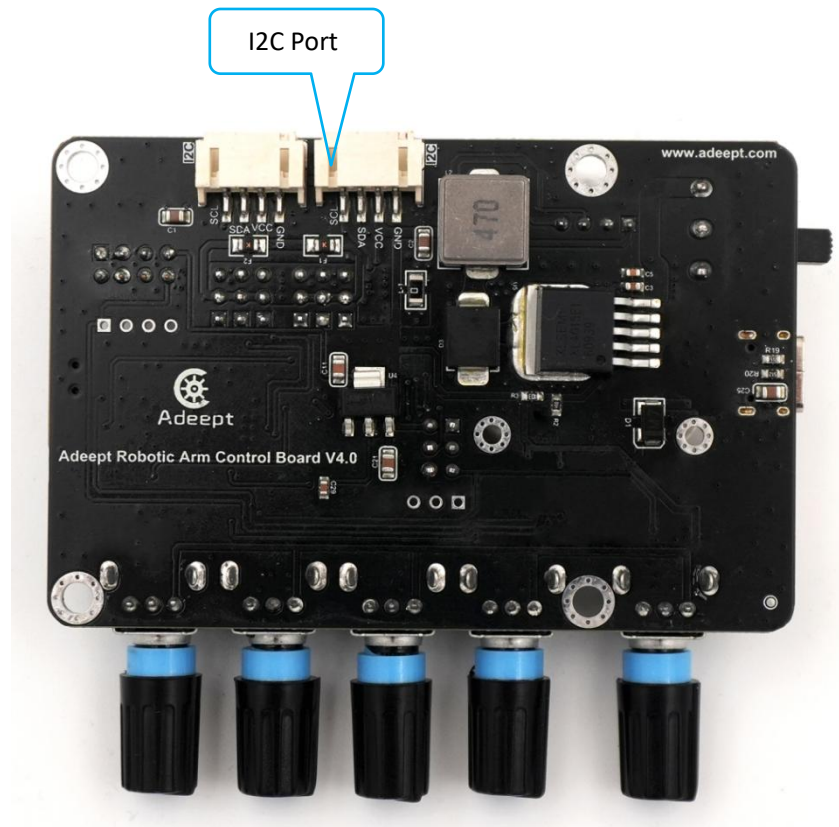


1 Introduction to the Adept Robotic Arm Control Board

1.1 Adept Robotic Arm Control Board

The Adept Robotic Arm Control Board development board is the main component of the Kit. Similar to the Arduino UNO development board, it is also an easy-to-use open source electronic prototyping platform, including the hardware part and the software part (Arduino IDE). The Adept Robotic Arm Control Board development board is mainly composed of a microcontroller (MCU), a universal input/output interface, etc. You can understand it as a microcomputer motherboard. We will introduce the Adept Robotic Arm Control Board development board in detail.





Name	Description
Type-C Port	Upload the program/charge the battery of the power interface
Power	Used to power the Adeept Robotic Arm Control Board, while the power indicator turns on simultaneously
RESET	Restart
Switch	When the USB cable is connected, the switch only controls power for the ESP8266 and servos. When the USB cable is disconnected and powered solely by battery, the switch controls the power supply for all components.

Servo	Servo interface controlled by Arduino pins
I2C Port	Used to connect I2C equipment, there are multiple I2C interfaces
WS2812 LED	2 WS2812 LED modules onboard
Buzzer	passive buzzer
ESP8266	Used to connect ESP8266 module.
Rotary Potentiometer	Rotary potentiometer, used to control the servo.
Upload Switch	Upload the program-0 ESP8266 Communication-1
IR Receiver	Receive infrared signals
Button	Programmable Button

1.2 Precautions for Using the Adeept Robotic Arm Control Board

The front and back sides of the Adeept Robotic Arm Control Board are equipped with many components. Please note to prevent short circuits when using it, and turn off the power before plugging in or unplugging cables to avoid damaging the control board.

The Adeept Robotic Arm Control Board requires two standard 18650 lithium batteries for power supply, and each battery has a standard voltage of 3.7V.

Regarding the charging function: the **CR** red light turns on during charging, and the red light turns off when fully charged.

Regarding the **Upload Switch**, please note that:

1. If you install the ESP8266 module to the Adeept Robotic Arm Control Board, you need to make sure that the paddle of the Upload switch is at **0** every time you upload the program.
2. Every time you use the ESP8266 module to communicate with the Adeept Robotic Arm Control Board, you need to ensure that the paddle of the Upload switch is at **1**.

